Validation Evidence

Hospital Operations, Cost & Patient Flow Intelligence | AnalystLab Africa Experience Lab, Week 3, Track A: Data Analytics | Prepared by Kukoyi Zainab Omotola

1. Purpose and method

This document records how the dashboard figures were checked against the source data. Each key figure was calculated independently in three ways and compared with the Power BI dashboard: directly from the source CSV files (profiling), with Excel formulas on the encounter and procedure data, and with SQL queries. The full comparison table is the Validation Values sheet in KPI_Definitions.xlsx, and the SQL is in validation_queries.sql.

A result is a Match when the difference from the source value is less than 0.005. Cards that the dashboard shows in rounded form (for example $101.5M) are checked by rounding the source value the same way.

2. Reconciliation results

Measure	Source file	Excel	SQL	Power BI	Status
Total Encounters	27,891	27,891	27,891	27,891	Match
Total Claim Cost	101,514,375.52	101,514,375.52	101,514,375.52	$101.5M (rounded card)	Match
Total Payer Coverage	31,097,506.99	31,097,506.99	31,097,506.99	$31.1M (rounded card)	Match
Unique Patients	974	974	974	974	Match
Procedure Count	47,701	47,701	47,701	47,701	Match
Avg Procedure Base Cost	2,212.06	2,212.06	2,212.06	$2,212 (rounded card)	Match
Zero Coverage Encounters	13,586	13,586	13,586	13,586	Match
Encounters Over 24h	75	75	75	75	Match
Avg Claim Cost per Encounter	3,639.68	3,639.68	3,639.68	$3,640 (rounded card)	Match
Payer Coverage %	30.63%	30.63%	30.63%	30.6%	Match
Zero Coverage %	48.71%	48.71%	48.71%	48.7%	Match

Measure	Source file	Excel	SQL	Power BI	Status
Over 24h %	0.269%	0.269%	0.269%	[confirm from Power BI card]	Excel and SQL match source

The three independent checks required by the assignment are Total Encounters, Total Claim Cost and Zero Coverage Encounters. All three agree across the source file, Excel, SQL and Power BI.

3. How each check was done

Excel. The encounter fields (Id, patient, start, stop, class, total claim cost, payer coverage) and procedure fields (encounter, description, base cost) were placed in Enc_Data and Proc_Data. The Excel Calc sheet uses COUNTA, SUM, AVERAGE and COUNTIF to reproduce each figure. Duration is measured in seconds from START to STOP, so an encounter of exactly 24 hours is not counted as over 24 hours. A pivot table of encounter class against encounter count and total claim cost was also built [insert pivot screenshot].

SQL. Each figure has its own SELECT query in validation_queries.sql. Over 24 hours uses TIMESTAMPDIFF(SECOND, START, STOP) > 86400. Only SELECT statements were used, so no data was changed. [Confirm: queries run in the isolated practice database; insert result screenshots.]

4. Filter test

The totals were tested with the Encounter Class slicer set to each class. The dashboard class table was compared with the source data, grouped by class.

Encounter class	Encounters	Total claim cost	Avg claim cost	Over 24h	Dashboard vs source
ambulatory	12,537	$36,283,440.84	$2,894.11	5	Match
emergency	2,322	$10,750,051.53	$4,629.65	5	Match
inpatient	1,135	$8,809,134.34	$7,761.35	58	Match
outpatient	6,300	$14,094,967.12	$2,237.30	7	Match
urgentcare	3,666	$23,349,349.96	$6,369.16	0	Match
wellness	1,931	$8,227,431.73	$4,260.71	0	Match
All classes	**27,891**	**$101,514,375.52**	**$3,639.68**	**75**	**Match**

The six classes add up exactly to the unfiltered totals (27,891 encounters, $101,514,375.52 and 75 encounters over 24 hours), so the figures change correctly under filtering. [Insert screenshot of the dashboard filtered to inpatient.]

5. Discrepancies investigated

No discrepancy between the dashboard and the source data was found. One point was examined: Excel's unrounded average procedure base cost is 2,212.0650, against 2,212.06 in the source profile. This is a rounding difference of 0.005, not a calculation error.

6. Metrics that could be misleading

- **Over 24h %.** The metric counts encounters of strictly more than 24 hours (75). Another 1,077 encounters last exactly 24 hours and are excluded. A rule of 24 hours or more would give 1,152. The 75 also include very long records (13 exceed 30 days), so the metric should not be read as a measure of hospital stays.
- **Zero Coverage %.** A payer coverage of 0 may mean uninsured, unrecorded or unpaid care, and the data does not say which. It covers all NO_INSURANCE encounters and some with named insurers.
- **Avg Claim Cost per Encounter.** A change in the mix of encounter classes can move the average without any change in prices.
- **Procedure measures.** 13,221 encounters (47.4%) have no procedure record, so procedure figures cover only part of the activity. Base cost is not claim cost, and the two should not be added or compared directly.
- **Currency.** The data does not state a currency, so $ is used as a display symbol only.

7. Limitations of this validation

The source-file values were calculated from the same CSV files that feed the dashboard, so these checks confirm that the dashboard calculates correctly from the data supplied. They do not prove that the data itself is accurate. The data-quality findings in the Data Quality Assessment document apply.

8. Tools and acknowledgement

Power BI, Power Query and DAX (dashboard), Excel (formula and pivot checks) and SQL (queries). AI assistance (Claude) was used for planning, profiling and documentation, and all results were reviewed by the author.